Güngör Deniz Karataş

Full-Stack Engineer | TypeScript, Python, Node.js | AI Agents & LLM Applications

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SUMMARY

Full-stack software engineer with 5+ years of experience building web applications, REST APIs and data-heavy business systems with Node.js, NestJS, TypeScript, Angular and SQL. Owned backend architecture, databases and on-premises deployments in a small team, working directly with clients. Recently built a tool-calling AI agent for bank transaction review (Python, Google ADK, RAG) with deterministic controls, human review and an evaluation suite.

SKILLS

Backend: Node.js, NestJS, Express, REST APIs, WebSocket (Socket.IO), JWT, OAuth 2.0, RBAC, multi-tenant SaaS, scheduled jobs

Frontend: Angular (Signals, standalone components, NgRx, lazy loading), RxJS, TypeScript, JavaScript, HTML, CSS, SCSS

Databases: SQL, Oracle, MySQL, PostgreSQL, Redis, Prisma, SQLite, MongoDB

AI: LLM APIs (OpenAI, Gemini), AI agents, tool calling, RAG, vector databases (ChromaDB), Google ADK, streaming, agent evaluation

DevOps & Testing: Docker, Docker Compose, CI/CD (GitHub Actions), Linux, Nginx, PM2, IIS, Jest (unit, integration, e2e), pytest, Git, Nx monorepo

Languages: TypeScript, JavaScript, Python, SQL

EXPERIENCE

Independent Software Engineer | Own products, Remote — Jun 2025 – Present

Bank Agent Lab: AI transaction risk review agent (banking domain) | github.com/GDenizKaratas/bank-agent-lab

Python, Google ADK, Gemini, RAG, ChromaDB, SQLite, Streamlit, pytest

- Built an AI agent that investigates suspicious transactions through four scoped, read-only tools.
- Kept risk decisions in deterministic Python checks, separate from LLM reasoning, and used RAG over policy documents to ground explainable risk reports.
- Added human review, audit logs and a repeatable evaluation set; 120+ automated tests cover core workflows and controls.

Content Pilot: AI social media scheduler (multi-tenant SaaS) | github.com/GDenizKaratas/nest-ai-pilot

NestJS, TypeScript, Prisma, MySQL, OpenAI API, OAuth 2.0, Angular

- Built a multi-tenant NestJS backend with JWT auth, role-based access control, team invitations, plan-based quotas and a relational schema of 30+ Prisma models.
- Integrated OAuth (X with PKCE, LinkedIn, Instagram) with scheduled token refresh, and publishing of text, images and Instagram carousels, immediate or scheduled.
- Added streamed AI content generation with per-plan quotas, Iyzico subscription billing, S3-compatible media storage (Cloudflare R2) and CI on GitHub Actions.

Full-Stack Engineer | ARTEON Software Consultancy, Istanbul — Feb 2020 – Jun 2025

- Owned backend architecture, database design and deployments across internal and client projects in an Nx monorepo, working directly with clients.
- Built the core of a no-code application platform: a visual data-model designer and a Node.js service that generates JWT-protected CRUD APIs from user-defined models.
- Built a reporting and analytics engine that runs dynamic, user-defined SQL queries on Oracle and MySQL, with Redis caching and Angular dashboards.
- Developed MES and ERP integrations that move production data between factory systems and business workflows.
- Built a project management platform with an interactive Gantt chart in Angular, real-time task tracking and a REST API backend.
- Built scheduled background jobs for automation, monitoring and automated reporting, reused across client projects.
- Started the social media scheduling platform and its third-party API integrations, later continued independently.
- Deployed and maintained applications on customer-managed Linux and Windows (IIS) servers with Nginx, PM2 and Docker in isolated on-premises environments.

EDUCATION

B.Sc. Management Information Systems | Anadolu University — 2025 – Present

B.Sc. Textile Engineering (100% English) | Istanbul Technical University — 2014 – 2019

CERTIFICATIONS & LANGUAGES

IBM Full-Stack Developer Course (2024 – 2025) | Turkish (native) | English (C1; 6-month advanced program at ILSC Sydney)